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Ruffed Grouse *Bonasa umbellus* habitat selection in a spatially complex forest: evidence for spatial constraints on patch selection

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Drumming display sites of male Ruffed Grouse *Bonasa umbellus* serve as activity centres and occur in a variety of forest types and age classes. We used a case-control study design and model selection to evaluate a set of predictions about habitat constraints placed upon Grouse given the habitat where their breeding season activity centres were located. We located 110 individual Grouse activity centres near Cloquet, Minnesota, between 2002 and 2005, and 40 activity centres at two other sites in northern Minnesota in 2005. Our most parsimonious model indicated that Ruffed Grouse used logs more than other potential drumming structures (e.g. stumps, dirt mounds or roots) and sites with a higher density of shrubs as compared with unused sites. Predicted values from this model correlated with observed values from independently sampled areas. Structures used by Grouse for their drumming display were characterized by a greater density of aspen stems than unused sites when in young aspen and mature pine, but not when in other forest types. The patterns of habitat selection that we observed supported predictions (1) that the differences in habitats at larger spatial scales (e.g. forest stands or breeding territories) may impose certain constraints on specific sites selected for their drumming display, and (2) that some of these constraints may vary by forest type.

Keywords: case-control design, forest management, *Pinus*, *Populus*.

Habitat selection may follow a top-down or bottom-up process (Kristan 2006). Mobile species are likely to follow a top-down process where individuals select home ranges within landscapes, habitat patches within home ranges, and then specific sites (e.g. foraging or nesting sites) within patches (Hildén 1965, Johnson 1980). Alternatively, individual birds may sample habitat from the bottom-up by occupying a specific site (e.g. nest tree) containing desirable attributes, regardless of surrounding habitat structure at a larger scale (Kristan 2006). Consequently, patterns of selection may be constrained by decisions made at different scales (Orians & Wittenberger 1991) and observed patterns of habitat selection at any given scale may be a result of selection for attributes at a different scale (Wiens 1989).

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Habitat selection studies are often based on comparisons between characteristics at used and unused sites (Manly *et al.* 2002, Keating & Cherry 2004, Anderson & Gutzwiller 2005). Habitat studies that investigate selection at different scales separately may be difficult to interpret because observed patterns may be confounded by correlations among habitat attributes selected at different scales (i.e. cross-scale correlations; Battin & Lawler 2006). Comparing habitats at used and unused sites where sampling of unused sites is conditional on some criteria (e.g. unused sites matched to used sites by distance or forest type) is one approach that explicitly accounts for selection at larger spatial scales. For example, a comparison of habitat between used and unused sites where each unused site is randomly selected within an individual's home range controls for constraints imposed on smaller-scale selection (e.g. patch selection within a home range) by larger scale decisions (e.g. selection of the home range; Jones 2001, Battin & Lawler 2006). Study designs where matching is conducted with respect to response variables, when an unused site is sampled to compare habitat with a used site within an individual's home range, are referred to as case-control designs (Agresti 1996).

Ruffed Grouse *Bonasa umbellus* are gallinaceous birds that show habitat correlations at different scales (Rusch *et al.* 2000, Whitaker *et al.* 2006). This makes them a model species for examining questions of small-scale habitat constraints imposed by habitat at a larger scale. For example, male Ruffed Grouse are associated with aspen (*Populus* spp.) stands throughout much of their range (Gullion & Alm 1983, Rusch *et al.* 2000). In addition, they select particular elevated structures as breeding display sites which are surrounded by dense woody vegetation within these stands (Gullion & Alm 1983, Rusch *et al.* 2000).

Female Ruffed Grouse visit males that display in dense woody vegetation during the breeding season, and later use these habitats to rear their young. Thus, the display site and surrounding vegetation are important components of Ruffed Grouse habitat (Rusch *et al.* 2000). Displaying male Grouse produce a non-vocal sound, called 'drumming', by rapidly beating their wings in the air while perched on an object higher than ground level such as a log, rock or root mass (Rusch *et al.* 2000). The primary structure where the drumming display occurs represents the territory centre of a male Grouse during the spring breeding period (Aubin 1972, Boag 1976). Therefore, vegetation structure surrounding display structures is critical Ruffed Grouse habitat because the birds are vulnerable to predation when they perform their drumming display (Rusch *et al.* 2000). The nature of habitat structure surrounding drumming display sites has been the focus of many studies (Gullion *et al.* 1962, Palmer 1963, Gullion 1967, Boag & Sumanik 1969, Rusch & Keith 1971, Boag 1976, DeStefano & Rusch 1984). However, none has either directly assessed the relative importance of various habitat characteristics on drumming site occupancy or accounted for the influence of habitat at other scales on drumming site selection. This lack of scale-dependent analysis has led to disagreement about the relative importance of small-scale habitat characteristics to Ruffed Grouse (Rusch & Keith 1971, Boag 1976, Rusch *et al.* 2000). For example, Gullion and Alm (1983) suggested that Ruffed Grouse prefer aspen at the scale of forest stands, but Gullion (1990) stated, without explicit examination, that '... under some conditions high densities of breeding Ruffed Grouse may exist in association with pine plantations'. In our study population, most Ruffed Grouse selected young (10–25 years of age) aspen stands during a period of low population density (i.e. at the bottom of their 10-year cycle; Zimmerman 2006). However, we observed some individuals using other forest types (e.g. *Pinus* sp.) where individuals presumably had lower relative fitness (Zimmerman 2006), when theory predicted (*sensu* Fretwell & Lucas 1970) that they should have occupied only the best habitats (aspen). The inconsistency between observed habitat selection and theoretical prediction led us to investigate the question: why did Grouse occupy forest stands that were apparently suboptimal? Therefore, we assessed habitat selection for drumming display sites at a small scale (< 10 m), which was conditional upon birds having selected breeding territories at a larger scale (i.e. the forest stand, which had a mean ± sd area of 2.48 ± 3.95 ha at Cloquet). Specifically, we used a case-control design and model selection to compare the relative importance of various habitat conditions for site selection by Ruffed Grouse within their breeding season territories as a way to assess possible constraints placed upon the birds given the forest type within which the display sites occurred.

METHODS

Study area

We studied Ruffed Grouse at three locations throughout northern Minnesota, USA: Cloquet Forestry Center (hereafter Cloquet); Chippewa National Forest (hereafter Chippewa), and Cass and Hubbard counties (hereafter Cass). We established nine survey transects at Cloquet that provided a complete survey of all forested patches within the study area from 2002 to 2005, whereas we randomly selected three transects within each of the other two areas, which we surveyed in 2005 (Zimmerman & Gutiérrez 2007).

Cloquet (46°31′N, 92°30′E) was a research forest managed by the University of Minnesota, and was located approximately 30 km west southwest of Duluth, Minnesota. The centre of the Cass area (46°56′N, 94°31′E) was approximately 150 km west of Cloquet. Chippewa (47°14′N, 93°32′E) was approximately 105 km northwest of Cloquet. Surveys at Cass occurred on public lands managed by local county governments, whereas surveys within Chippewa occurred on public land managed by the

United States Forest Service. Cass and Hubbard counties and the US Forest Service managed their lands for multiple uses. Cloquet, Cass and Chippewa were in the transitional zone between the boreal and broadleaved deciduous forest biomes (Bailey 1995). When considering the 2005 data and individuals within 175 m of transects only, mean (± sd) drumming Grouse densities were similar among the three areas ($\bar{x}_{\text{Cloquet}} = 0.04 \pm 0.03$ ha^{-1}, $\bar{x}_{\text{Cass}} = 0.03 \pm 0.01$ ha^{-1} and $\bar{x}_{\text{Chippewa}} = 0.03 \pm 0.02$ ha^{-1}). Winters were cold and dry, while summers were warm and humid.

Almost 65% of Cloquet was upland forests and clearings, whereas the remainder was wetlands. Because Cloquet was a research forest, many upland forest stands were experimentally manipulated, resulting in a complex mosaic of forest types, age classes, stand structures and other vegetation types (e.g. open fields). Pine (mostly Red Pine *Pinus resinosa* and Jack Pine *P. banksiana*) and aspen (dominated by *Populus tremuloides*, but with some *P. grandidentata*) were the most common forest types, covering 28 and 16% of Cloquet, respectively. Forest stands of White Spruce *Picea glauca*, Balsam Fir *Abies balsamea*, Paper Birch *Betula papyrifera* and northern hardwoods (i.e. a mix of aspen, Paper Birch, Red Maple *Acer rubrum* and Red Oak *Quercus rubra*) also occurred in the areas but were not common. Beaked Hazel *Corylus cornuta* was the dominant shrub species throughout the study area. Other common shrubs included Juneberry *Amelanchier* spp., Pin Cherry *Prunus pennsylvanica*, Chokecherry *Prunus virginiana* and alder *Alnus* spp. Ruffed Grouse inhabited most forest types, particularly aspen, which they used in most age classes and densities. Vegetation at Cass and Chippewa was similar to that of Cloquet and was typical of habitats near the transition zone between boreal and northern hardwood forests.

Grouse surveys

We surveyed Ruffed Grouse along transects, located their drumming display sites and sampled micro-site habitat characteristics surrounding their drumming structures. The sampling units for our analysis were small areas (see below) centred on the physical structures used by Grouse for drumming displays and randomly selected unoccupied structures found within breeding season home ranges. We located these structures by listening for the drumming displays made by males during morning surveys along nine permanently marked transects at Cloquet. We established these transects such that no upland forests were > 175 m from any transect. The design of these transects allowed unbiased estimates of site occupancy by drumming males within 175 m of transects (Zimmerman & Gutiérrez 2007), which resulted in an unbiased estimate of male Grouse habitat use at Cloquet. We began surveys 30 min before sunrise on days when winds were < 7 m/s. Surveys lasted for 2–4 h and depended on the number of Grouse detected and the time taken to locate drumming structures. During surveys, we walked transects while listening for drumming Grouse. Once a male was heard, we located the display structure, recorded the structure's location by estimating its Universal Transverse Mercator (UTM) coordinates using a Global Positioning System (GPS) and marked the structure with a uniquely numbered metal tag (Zimmerman & Gutiérrez 2007). We sampled each transect 15 times during the spring season (Zimmerman & Gutiérrez 2007). We used mirror traps to catch Grouse (Gullion 1965), which we ringed with uniquely numbered aluminium leg rings. To ensure the unique occupancy by an individual Grouse of each drumming site, we attempted to capture and ring each Grouse or note timing and consistency of drumming locations. We used the same field techniques and data collection protocols at Cass and Chippewa, with the exception that we did not trap and ring Grouse in those areas.

Habitat data

We measured the characteristics of the drumming structure and the forest stand immediately surrounding the drumming structure at each occupied and randomly selected unused site. We used a case-control design to select unused sites at the three study areas (Cloquet, Cass and Chippewa) to deal with two important issues when assessing habitat use. First, as discussed earlier, habitat selection by animals at various scales could lead to cross-scale correlations. Secondly, habitat abundance does not always indicate habitat availability because some habitats may exist, but are unavailable to individuals because of predators, competitors or other factors (Jones 2001). Therefore, we conditioned our sampling such that randomly selected unused sites had to occur within an area equal to the size of an average breeding-season home range of a male Ruffed Grouse (2.3 ha; Archibald 1975). Thus, our analysis addressed the following question: given the forest habitat within an individual's territory, how was forest structure different between the used drumming location and a randomly selected, and available, location within its territory? Our case-control sampling design also addressed the prediction that habitat constraints at small scales (e.g. habitat within 10 m of drumming structures) could be important if individuals occupy suboptimal or less-preferred forest stand types. We used ArcView 3.2 (ESRI Inc., Redlands, CA, USA) and a random point generator (Jenness 2005) to randomly select unused available sites. We chose the nearest log, stump, root mass or dirt mound to the randomly selected point to represent an unused, but available, drumming structure. We searched each unused structure for droppings and feathers to ensure that they were not occupied by drumming Ruffed Grouse.

We recorded the type (log, stump, rock, dirt mound, root mass) of each structure used as a drumming display site, and the height and diameter of each drumming stage, which was the place on the structure where the individual Grouse perched while drumming. We also recorded the direction, distance from the drumming display stage and diameter of a 'guard object' (e.g. tree stem, bush or root flare, which could provide protection from a predator approaching from behind a drumming male; Gullion 1984). We used fixed-radius plots of 3.6 m to estimate stem density, by species, of shrubs (woody vegetation < 5 cm in diameter) and trees (woody vegetation ≥ 5 cm in diameter) > 0.6 m above drumming structures. We used variable probability sampling to quantify tree density to make our results comparable with data gathered by forest managers (Husch *et al*. 2003). We used wedge prisms to sample trees within variable plots and recorded the diameter at breast height (1.37 m) of each tree we used for estimating basal area. We estimated the density of woody stems in two ways: (1) by counting woody stems (trees and shrubs pooled) within the 3.6-m fixed-area plots surrounding drumming logs; and (2) by counting the number of trees using variable plots, which we converted to densities using standard formulas (Husch *et al*. 2003). We converted both of these variables to numbers per ha, and then subtracted the density of trees per ha from the density of stems per ha to estimate the density of shrubs per ha.

Data analysis

Analytical approach

We used model selection based on conditional likelihood to compare hypotheses about possible constraints that selection of breeding-season home ranges may impose on the selection of particular drumming structures by Ruffed Grouse. We developed 11 *a priori* hypotheses based on a literature search, field observations and an all-possible variable combinations modelling of unpublished data on Grouse habitat collected by G.W. Gullion at Cloquet between 1980 and 1989. These hypotheses were developed to represent a suite of conditions that would reflect an evaluation of the ecological consequences for a bird's selection of drumming sites conditional on forest type surrounding display sites. We expressed these hypotheses as statistical models (Table 1) and used a small sample size adjustment of Akaike's Information Criteria (AIC_c) to rank these and identify the most parsimonious model (Burnham & Anderson 1998). We estimated that the probability of detecting a Grouse on at least one of the 15 surveys conducted each spring was > 0.997 (Zimmerman & Gutiérrez 2007), so our sampling adequately determined which sites were occupied by Grouse each spring. We used the habitat variables described above as predictor variables for each model and conducted a correlation analysis to ensure that variables in the same model were not correlated. All predictor variables within the same model had Pearson correlation coefficients < 0.40. After we identified the best model with AIC_c, we used the data collected at Cass and Chippewa to assess the predictive ability of this model.

A priori hypotheses

We refer to the factors used in models as 'constraint factors' because they may constrain Grouse selection of home-range characteristics at a larger scale (forest stand). Furthermore, we recognized two classes of constraint factors necessary for site occupancy by Ruffed Grouse. The first class represents variables associated with a specific drumming structure, such as structure type, height and the presence of a guard object (Table 1). The second class represents factors associated with the forest attributes surrounding drumming structures, including shrub and tree density, and tree basal area, all of which provide important protection from predators (Table 1). Zimmerman (2006) presents a detailed discussion and supporting literature for each *a priori* hypothesis.

Estimating model parameters

We used a case-control design with logistic regression (Agresti 1996) to estimate the parameters within *a priori* models relating the effects of drumming

Table 1. Predictions of the effect of habitat on the occupancy of drumming logs by Ruffed Grouse at the Cloquet Forestry Center, Minnesota, 2002–05.

Model no.	Model description	Model variables* and predictions†
1	Drumming structure type	$+L$
2	Drumming stage diameter and height	$+DS_{Dia} + DS_{Ht}$
3	Guard object	$+GO_{Dia} - GO_{Dist} + GO_{Dir}$
4	Stem density – All types and species pooled	$+ST$
5	Aspen stem density, plus conifer stem density, plus shrub density	$+A_{ST} \pm C_{ST} + SH$
6	Aspen tree, plus other tree, plus shrub density	$+A_{TR} \pm O_{TR} + SH$
7	Edge avoidance	$\pm ED_{Index}$
8	Distance to nearest mature aspen	$-MA_{DIST}$
9	Distance to nearest mature pine	$+MP_{DIST}$
10	Combined model nos. 1 and 5	$+L + A_{ST} \pm C_{ST} + SH$
11	Combined model nos.1, 4 and 9	$+L + ST + MP_{DIST}$

*A_{ST}, density (no./m^2) of aspen stems (shrubs and trees pooled) > 0.6 m above drumming stage; A_{TR}, density (no./m^2) of aspen trees (woody vegetation > 5 cm dbh) surrounding drumming structures; C_{ST}, density (no./m^2) of conifer stems (pine, spruce and fir shrubs and trees pooled) > 0.6 m above drumming stage; L, drumming structure type: 1 = log, 0 = root, stump, dirt mound; DS_{Dia}, diameter (cm) of drumming structure at the drumming stage; DS_{Ht}, height (cm) of drumming structure at drumming stage; ED_{Index}, edge index – calculated by dividing the distance (m) to nearest edge by the forest stand size (ha); GO_{Dia}, diameter (cm) of guard object measured at a height of 0.75 m above drumming logs; GO_{Dir}, relative direction (range = 0–90°) of guard object from drumming stage: values close to 90° = guard object directly behind drumming stage, values close to 0° = guard object directly to left or right of drumming stage; GO_{Dist}, distance (m) from guard object to centre of drumming stage; MA_{DIST}, distance (km) from drumming structure to nearest mature aspen stand; MP_{DIST}, distance (km) from drumming structure to nearest mature pine stand; O_{TR}, density (no./m^2) of non-aspen trees (woody vegetation > 5 cm dbh); SH, density (no./m^2) of shrubs (woody vegetation < 5 cm dbh) > 0.6 m above drumming logs; ST, density (no./m^2) of woody stems (all species and sizes pooled).
†+, predicted positive correlation; –, predicted negative correlation; ±, strong correlation: could be positive or negative.

display habitat to site occupancy. We assumed a binomial response with used sites assigned a value of 1 and unused sites assigned a value of 0. The general form of the logistic models was:

$$\mathrm{logit}(\phi) = \beta * \Delta X.$$

The model did not include an intercept and the $\beta * \Delta X$ was a design matrix of habitat variables that could possibly distinguish between used and unused activity centres, where β represented unknown parameters to be estimated and the ΔX represented the difference in habitat metrics of interest at used and unused sites (Allison 1999). Therefore, each matched pair consisted of one data entry: the difference in the X metrics of interest as independent variables and the difference of responses (i.e. $1 - 0 = 1$) as the dependent variable (Agresti 1996). We estimated the unknown β parameters and model fit using conditional maximum likelihood (PROC LOGISTIC; SAS Institute 2004). Although the analysis of data using a case-control logistic regression provides an effective technique for analysing conditional habitat use (i.e. each randomly selected unused site is conditional on the animal selecting the used site), the dependency between the used and unused sites requires special consideration when interpreting results (Keating & Cherry 2004). The estimated parameters cannot be used to predict occupancy of drumming structures, but can be used to estimate the relative differences in habitat metrics at used versus unused sites (Keating & Cherry 2004). Therefore, β parameter estimates > 0 suggest that a particular habitat metric is greater at used sites relative to unused sites, values < 0 suggest that the metric is less at used sites compared with unused sites, and values of zero indicate that the metric does not differ between used and unused sites. We used Wald 95% confidence intervals (CI) to assess whether the β parameters were significantly different from 0.

Model selection provides a relative (i.e. relative to the other models in the *a priori* set) ranking of hypotheses, so we assessed the absolute performance of the best model in two ways. First, we estimated an R^2 value to assess how much variation in the data was explained by each model. Because R^2 for logistic regression is < 1 (Nagelkerke 1991), we estimated a maximum rescaled R^2 ($\bar{R}^2$; SAS Institute 2004). Nagelkerke (1991) defined the maximum rescaled R^2 as:

$$\bar{R}^2 = \frac{R^2}{\max(R^2)} \text{ with } \max(R^2) = 1 - L(0)^{2/n}$$

where $L(0)$ is the likelihood of a null model. We also assessed model performance using data collected at Cass and Chippewa, which were independent of model selection and parameter estimation. If our best model from Cloquet was accurate, the estimated β parameters and observed habitat differences at used and unused sites (i.e. ΔX at Cass and Chippewa) should discriminate between the used and unused sites sampled at Cass and Chippewa. Therefore, if the observations at Cass and Chippewa supported our best model from Cloquet, our estimated Φ should be above 0.50. If our observed habitat metrics did not distinguish between used and unused sites or showed the opposite pattern to our best model prediction, then the back estimated Φ would be = 0.5 or < 0.5, respectively.

Although Ruffed Grouse appeared to select aspen in northern Minnesota, they still used other forest types (Zimmerman *et al.* 2007). We assessed whether site-level constraints varied by forest type by estimating the model parameters from the best model separately for young aspen, mature aspen and mature pine because these forest types had the highest counts of drumming males. We included a fourth 'stand type' by pooling the remaining drumming structures (i.e. those in mixed northern hardwoods, spruce-fir, young pine). Estimating forest-specific parameter estimates reduced our sample size for this component of the analysis. To increase power in detecting evidence of forest-type-specific correlations, we presented 90% Wald confidence intervals with parameter estimates to assess whether individual variables are particularly important in certain habitats.

RESULTS

Grouse surveys

We located 110 independent Ruffed Grouse activity centres at Cloquet from 2002 to 2005. Grouse occupied some activity centres for > 1 year, and used different primary drumming structures within some activity centres among years (total number of primary drumming structures located = 209). The number of occupied activity centres declined throughout the study ($n = 64$ in 2002, $n = 51$ in 2003, $n = 48$ in 2004, $n = 46$ in 2005). We found grouse drumming display structures in most forest types, with most of the drumming sites in our study occurring in young (10–25 years old) aspen (45%), mature aspen (21%) and mature pine (15%) stands. Only 12% of the activity centres were occupied during all 4 years of the study, whereas 47, 26 and 15% of the activity centres were occupied for 1, 2 and 3 years, respectively. We located 19 and 21 individual territorial Grouse at Cass and Chippewa, respectively, in 2005.

Model selection and testing

Our *a priori* model that included the combination of drumming structure characteristics and surrounding forest stand attributes provided the best fit to the data (Table 2). This model indicated that grouse selected logs over other physical structures (e.g.

Table 2. Results of model selection for *a priori* hypotheses of drumming Ruffed Grouse habitat use, Cloquet Forestry Center, Minnesota, 2002–04

Model no.	Model description	–2 * Log Like	K*	AIC$_c$	AIC$_c$ weight	$\bar{R}^2$†
10	Combined model nos. 1 and 5	181.85	4	190.04	0.60	0.54
11	Combined model nos. 1, 4 and 9	184.71	3	190.83	0.40	0.53
1	Drumming structure type	221.45	1	223.47	0.00	0.37
2	Drumming stage diameter and height	248.28	3	254.39	0.00	0.35
3	Guard object	224.86	3	230.98	0.00	0.36
6	Aspen tree, plus other tree, plus shrub density	226.78	2	230.84	0.00	0.24
5	Aspen stem, plus conifer stem, plus shrub density	250.30	3	256.42	0.00	0.23
4	Stem density – All types and species pooled	254.69	1	256.71	0.00	0.21
7	Edge avoidance	286.72	1	288.74	0.00	0.02
9	Distance to nearest mature pine	287.98	1	290.00	0.00	0.01
8	Distance to nearest mature aspen	288.20	1	290.22	0.00	0.01

*Number of parameters.
†Maximum rescaled R^2.

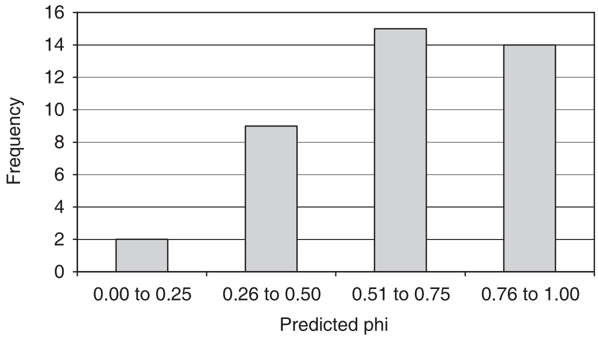

Figure 1. Histogram representing AIC_c best model discrimination between used and unused Ruffed Grouse drumming sites using data collected at Cass and Chippewa, USA, in 2005. Estimated phi = 1 represents perfect discrimination, values > 0.5 indicate some predictive ability, values = 0.50 indicate no predictive ability and values < 0.5 indicate that relative habitat conditions at used and unused sites are opposite that predicted.

stumps, roots, dirt mounds, rocks; $\hat{\beta}_{log} = 2.72$, 95% CIs = 1.85, 3.60). Grouse used logs more frequently (95% of drumming structures) than either mounds of dirt (4% of drumming structures) or stumps (1%) as drumming display sites. Approximately 66% of randomly selected unused structures were logs, 24% were stumps and 10% were dirt mounds. In addition, used sites were surrounded by a higher density of shrubs ($\hat{\beta}_{shrubs} = 0.50$, 95% CIs = 0.31, 0.69) than randomly selected unused sites. Although the model included aspen and conifer stem density, the correlation between these variables and occupancy was weak ($\hat{\beta}_{aspen} = 0.89$, 95% CIs = –0.59, 2.38; $\hat{\beta}_{conifer} = 1.10$, 95% CIs = –0.97, 3.18). This model explained approximately 54% of the variation ($\bar{R}^2 = 0.54$). The second ranked model, which was also based on a combination of log and surrounding vegetation variables, performed almost as well as the top model. This model also suggested a strong relationship between the presence of a log ($\hat{\beta}_{log} = 2.73$, 95% CIs = 1.85, 3.61) and used drumming sites. In addition, this model suggested that used sites had a higher density of stems, pooled across species, than unused sites ($\hat{\beta}_{stems} = 0.48$, 95% CIs = 0.29, 0.67). Although the distance to the nearest pine stand was included in this second-ranked model, this variable did not appear to differ between used and unused sites ($\hat{\beta}_{pine\ distance} = 0.002$, 95% CIs = –0.01, 0.01). Comparison of the best model to the data collected at 40 drumming territories at Cass and Chippewa suggested that this model performed reasonably well as the estimated Φ was > 0.54 at 73% (29/40) of the territories (Fig. 1). Although the model could distinguish used from unused sites at most of the Cass and Chippewa territories, the estimated Φ was 0.50 at one territory, which indicated that the model could not distinguish between the used and unused site, and < 0.50 (i.e. habitat conditions at used sites relative to conditions at unused sites were opposite what the model predicted) at 25% of the territories.

Table 3. Parameter estimates by forest type from the AIC_c best model for comparing habitat at used versus unused Ruffed Grouse drumming sites, Cloquet Forestry Center, Minnesota, USA, 2002–05

			90% CI	
Parameter	Forest type	Estimate	Lower	Upper
Aspen stem density	Young aspen	2.36	0.12	4.61
Aspen stem density	Mature aspen	–6.06	–12.36	0.24
Aspen stem density	Mature pine	8.50	1.17	15.83
Aspen stem density	Other	–1.47	–5.55	2.62
Shrub density	Young aspen	0.96	0.56	1.37
Shrub density	Mature aspen	0.48	0.15	0.81
Shrub density	Mature pine	0.52	0.14	0.91
Shrub density	Other	0.28	–0.11	0.68

Regardless of forest type, logs rather than stumps, dirt mounds or other structure types were present at used sites more than unused sites, and the density of conifer stems did not appear to differ between used and unused sites. Aspen stem densities appeared to be slightly higher at used sites than at unused sites in both mature pine and young aspen stands, but did not differ between used and unused sites in the other forest types (Table 3). The density of shrubs (woody vegetation < 5 cm diameter at a height of 1.37 m) was greater at used sites than unused sites in all forest types except the 'other' habitat type category, which was largely mixed hardwoods and spruce-fir (Table 3).

DISCUSSION

We assessed factors that might constrain Ruffed Grouse to use particular drumming sites within their breeding-season territories, particularly when occupying sites that we thought were suboptimal during a period of low population density. The conditional-sampling design (*sensu* Battin & Lawler 2006) that we used is an effective technique for assessing habitat use at small spatial scales because it controls for habitat at larger scales. This design is particularly germane to the study of Ruffed Grouse habitat selection because the species occupies a variety of forest types across most of boreal and north temperate North America. Many investigators have assessed habitat selection by Ruffed Grouse without directly comparing the relative influence of various habitat variables on site occupancy, or controlling for selection at other spatial scales, which has led to controversy about the relative importance of both small- and large-scale habitat selection in this species (Bump *et al.* 1947, Gullion 1967, 1990, Rusch & Keith 1971, Boag 1976, Rusch *et al.* 2000). Whitaker *et al.* (2006) assessed stand-type selection by Ruffed Grouse while controlling for landscape habitat in the Appalachian region of the eastern US and observed that the relative importance of forest stand types varied by landscape composition. At a smaller spatial scale, we found three factors that could be important constraints on selection of specific drumming display sites by Ruffed Grouse: (1) logs, (2) shrub density and (3) aspen stem density. In the first case, logs not only provide an elevated display stage, but are likely to have guard objects. They may also provide a more suitable platform for other types of displays (e.g. strutting). In the second and third cases, a high density of shrubs or stems probably provides the protective cover needed for Grouse when they are engaged in their drumming display. Furthermore, these habitat constraints appeared to vary by forest type surrounding display sites.

The availability of a log for the drumming display appeared to be a basic constraint for a site to be used by a drumming Ruffed Grouse during the breeding season. Although many potential drumming structures (logs, dirt mounds, stumps) are common in our study area, drumming males showed a particularly strong affinity for logs over other possible drumming structures. Gullion (1967) also observed that Ruffed Grouse often drummed on fallen trees or logs from felled trees that were left behind during timber harvests. However, he and others noted that Grouse also stand on a variety of other objects while drumming, including rocks, stumps, exposed roots, snow banks, mounds of dirt and piles of logs (Gullion *et al.* 1962, Rusch *et al.* 2000). Because potential drumming structures are common in forests and specific attributes of logs used by individual Grouse are similar to unused logs (Boag & Sumanik 1969), Rusch *et al.* (2000) hypothesized that the absence of drumming structures does not limit the distribution of Ruffed Grouse. Although we agree with this suggestion, we think that the absence of logs could limit the occupancy of particular stands by Grouse; 26 of 39 aspen stands were not occupied by Grouse annually, but seemed to have all the characteristics of other occupied stands except that they had few logs. Thus, we predict that the paucity of logs in a particular area could reduce Ruffed Grouse abundance locally.

A second constraint to occupancy of drumming sites appeared to be the density of shrubs (understorey woody vegetation < 5 cm in diameter) surrounding drumming structures. The density of shrubs was greater at drumming sites than at randomly selected unused sites in all but one forest type. Gullion *et al.* (1962) and Boag and Sumanik (1969) also found a similar positive effect of shrubs on drumming-log occupancy. However, Rusch and Keith (1971) reported that the density of shrubs surrounding drumming logs in Alberta was lower than at unused sites and concluded that the density of trees was more important than the shrub layer. Subsequently, Boag (1976) reported that Grouse abandoned logs where shrubs were experimentally removed. Our results support the importance of shrub cover to Ruffed Grouse in Minnesota. We suspect that pooling across multiple forest types confounded the estimation of the influence of shrubs and other habitat factors in the mixed northern hardwood/spruce-fir/young pine category. However, we did not assess patterns in more detail for this category because reclassification would further reduce sample size and power to detect any meaningful trends.

A third factor which may present a constraint for Ruffed Grouse selection of forest stands appeared to be the density of aspen stems (both understorey and overstorey classes pooled). After accounting for occupancy of young aspen stands in the case-control design, Ruffed Grouse appeared to use drumming structures in the more dense portions of those young aspen stands, as predicted by Zimmerman *et al.* (2007). Similarly, aspen stem density appeared to be important for individuals occupying mature pine stands. Although counts of drumming Grouse were relatively high in mature pine, this was the most common habitat in Cloquet and after accounting for habitat abundance this forest type was rarely occupied compared with young aspen (Zimmerman 2006). Gullion (1990) hypothesized that aspen patches as small as 0.4 ha are used by Grouse in pine plantations. We also observed very small inclusions being used by Grouse. Thus, the natural structure of pine stands may place a constraint on Grouse occupancy, which may be alleviated when small inclusions of aspen occur within pine stands. Future studies that quantify the size of inclusions and divide aspen stem density into overstorey and understorey categories would elucidate the nature of these aspen inclusions, as well as refine our constraint hypothesis for the selection of unlikely habitats.

We used data collected at Cloquet to derive model parameters and then test the best model with data collected at other sites. Although habitat proportions changed little among years, they differed among the sites (Zimmerman & Gutiérrez 2007), which could lead to differential cross-scale correlations. In addition, habitat selection may have changed among years because of fluctuating resources that we did not measure. Such changes could confound tests of models that spanned different temporal samples (Cloquet data = 2002–05, Cass and Chippewa = 2005 only). Hierarchical models (Kristan 2006) that include variables taken at different spatial and temporal scales and interactions between those variables would be a more explicit approach for developing our models. However, at this time, we do not have sufficient data to support such complex model structure. Nonetheless, our model performed reasonably well when compared with independent data we collected at Cass and Chippewa.

Zimmerman *et al.* (2007), using forest inventory data (tree density and basal area), noted that forest management for Ruffed Grouse should vary by forest type due to the influence of tree species composition and understorey structure. They noted that when Grouse occupied pine stands, they used ones that appeared to be managed for lower density and higher basal area of trees. They suggested that such stands may be characterized by higher shrub density and more aspen inclusions than stands managed for higher tree density because more sunlight will penetrate the open canopy of low-density stands. Our results appear to support their speculation concerning the mechanism behind predicted Grouse management of pine forests.

Regardless of the habitat selection process (i.e. top-down or bottom-up), the surrounding matrix of habitat at a larger scale may place constraints on small-scale site selection by avian species. Understanding habitat occupancy, the consequences of habitat choices and the interactions between the two is critical to improving our understanding of avian ecology (Kristan *et al.* 2007) and management (Anderson & Gutzwiller 2005). It may also help us to understand why some animals select habitats that may seem suboptimal, which has implications for conservation as well as evolutionary biology. While controlling for forest type, we demonstrated that habitats at a larger scale place constraints on microsite occupancy in the Ruffed Grouse. Estimation of survival rates, reproductive rates or female visitation rates among habitat types will provide valuable insight into the influence of these habitat selection constraints on the dynamics of Grouse, because we do not know if these aspen inclusions within pine stands are ecological traps (Gates & Gysel 1978).

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